**How to Avoide Overfitting:**

Avoiding overfitting in regression machine learning models shares some similarities with general techniques for avoiding overfitting, but there are also specific strategies tailored to regression tasks. Here are some techniques:

1. **Eda**
2. **For Regression model use Ridge(L2 Regularization)**
3. **Increase Training Data:**
4. **Cross-Validation:**
5. **User of Grid search CV, Random search CV**
6. **Feature Selection:**
    i. **Filter Method**
    ii. **Wrapper Mehod**
    iii. **Embeded Methods**
7. **Feature Extraction (Dimensionality Reduction)**
    i. **PCA**
    ii. **LDA**
    iii. **T-SNE**
8. **Feature Scaling:**
9. **Hyperparameter Tuning:**

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**Hyperparameter Tuning:**

Hyperparameters are settings or configurations of a machine learning model that are tuned by the practitioner to optimize the model's performance. Here are examples of hyperparameters for various machine-learning models:

**Increase Training Data:**

If possible, collect more training data to improve the model's ability to generalize. More data can help reduce overfitting by providing the model with a more representative sample of the underlying data distribution.

**Cross-Validation**:

Split your dataset into training, validation, and test sets. Use techniques like k-fold cross-validation to evaluate your model's performance on multiple splits of the data. This helps ensure that your model's performance isn't overly dependent on a particular subset of the data.

**Working with Feature**

**Feature Selection**:

**Choose** relevant features for your regression model and avoid adding noise or irrelevant features. Techniques like

- **Filter Method:**
- **Wrapper Method:**
- **Embeded Methods**
    a. **For linear and losgistic regression:**
        i. **Ridge regularization**
        ii. **Lasso Regularization**
    b. **Tree like algorythm:**
        i. **Early Stopping:**
        ii. **Preprunning**
        iii. **Post Pruning:**
        iv. **Ensemble Methods**

**Feature Extraction (Dimensionality Reduction) :**

It involves transforming raw data into a set of features that can be used for model training and evaluation. There are various techniques for feature extraction, depending on the type of data you're working with. Here's a rundown of some common methods:

**Dimensionality Reduction Techniques**

- **Principal Component Analysis (PCA)**:
- **Linear Discriminant Analysis (LDA)**:
- **t-Distributed Stochastic Neighbor Embedding (t-SNE)**:

**Feature Scaling**:

Normalize or standardize your features to ensure that they are on a similar scale. This can help prevent the model from being overly sensitive to the scale of different features.